



An effective strategy for degrading waste PET plastic into regenerated DMT with nearly equivalent methanol at relatively low temperature

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ABSTRACT

The increase in the usage of polyethylene terephthalate (PET) leads to the generation of a large amount of waste, which may cause carbon emissions and waste of petroleum resources. The methanolysis method has attracted global attention, but limited by high reaction temperature and excessive use of methanol. We report here an efficient catalyst, tetramethylguanidine/1,5-diazabicyclo[4.3.0]-5-non-ene (TMG/DBN), converting waste PET into regenerated dimethyl terephthalate (rDMT) under 100 °C with almost equivalent methanol, achieving 100 % PET conversion and 93.1 % yield of rDMT. A series of analytical methods and theoretical calculations were used to reveal the enhanced interaction between the catalyst and reactants, which facilitates the degradation of PET at low methanol dosages. In addition, a kinetic model was also developed with a relatively low activation energy of 84.9 kJ/mol. This work provides an effective strategy for the up-grading recycling of waste PET under mild reaction conditions.

1. Introduction

Plastics are widely used in daily life but cause significant environmental pollution and waste of resources facing the world today (Kim et al., 2024; Xie et al., 2024; Yang et al., 2024). Polyethylene terephthalate (PET), one of most commonly used thermoplastics, is applied in many fields such as fibers, bottles, packing materials and construction materials for its excellent properties (Huang et al., 2021; Wu et al., 2024; Zhang et al., 2024). The global PET market size is about \$43.22 billion in 2023 and will grow to \$67.13 billion in 2028 (Polyethylene Terephthalate Market Size, Share And Research Report, 2033). However, the increase in PET usage leads to the generation of a large amount of waste, which is landfilled or incinerated principally (Zhang et al., 2023a; Zhu et al., 2012). These methods not only leads to the waste of petroleum resources, but also has environmental impacts since emission of CO₂ (Guo et al., 2025; Jehanno et al., 2022; Vollmer et al., 2020; Zeng et al., 2024). Hence, effective strategies of PET recycling are needed for a

sustainable chemical industry and a carbon circular economy (Bai et al., 2023; Chen et al., 2023; Muangmeesri et al., 2024). Chemical recycling of PET is promising for its production of value-added chemicals (Dai et al., 2023; Li et al., 2022; Liu et al., 2023) and original monomers (Laldinpuui et al., 2022; Peng et al., 2023; Ügdüler et al., 2020; Zhang et al., 2023b), and consuming less energy and discharging less greenhouse gases.

In the chemical method, methanolysis of PET is prospective for large scale industrial production since the product, dimethyl terephthalate (DMT), is easy to separate and refine (Ye et al., 2023) and not affected by raw material source (Tollini et al., 2022), making it be used to produce recycled PET with high quality. DMT is also a precursor for the production of important chemicals like 1,4-cyclohexanedimethanol and dimethyl cyclohexanedicarboxylates (Huang et al., 2014; Xiao et al., 2022). Therefore, PET methanolysis is of great significance. However, the traditional methanolysis of PET usually takes place at high temperature, during 180 to 280 °C, and high pressure, during 2 to 4 MPa

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(Babaei et al., 2024), with the yield of DMT less than 90 % (Paszun and Szychaj, 1997; Pham and Cho, 2021). Methanolysis of PET in supercritical methanol progresses rapidly and realizes a high DMT yield (Liu and Yin, 2022) at high temperatures and pressures (Genta et al., 2005). However, the harsh reaction condition requires a much more complicated apparatus, which may increase industrial cost. Besides, in the process of PET methanolysis, with the progress of the reaction, the amount of by-product ethylene glycol (EG) is increasing, which leads to the transesterification of DMT and EG to 2-hydroxyethyl methyl terephthalate (MHET) and bis(hydroxyethyl) terephthalate (BHET) impurities as shown in Fig. 1(a) and Fig. 1(b), resulting in the waste of resource and increased costs for subsequent impurity removal. The chemical depolymerization of PET often introduces catalysts to accelerate the depolymerization rate. Currently, metal-based catalysts are used commonly (Du et al., 2020; Jiang et al., 2022; Tang et al., 2022; Ye et al., 2023). The reactivity of these catalysts is typically enhanced at elevated reaction temperatures but decreases rapidly at low temperature (Bai et al., 2023). Besides, the transesterification of DMT is observed to become more significant at elevated temperatures (Ganesan et al., 2009; Jadhav et al., 2020).

These years, researchers focus on mathanolysis at mild condition for the advantages of low carbon emissions and green energy-saving (Tang et al., 2022). Cosolvents are added into the system to enhance PET degradation. Tang et al (Tang et al., 2023) decline the PET methanolysis temperature to 120 °C with the addition of acetonitrile as cosolvent and the product yield is more than 80 %. Pham and Cho (Pham and Cho, 2021) use dichloromethane as a cosolvent with 22 times the mass of PET to obtain complete methanolysis at the temperature range of 25–30 °C within 24 h. The employment of cosolvents can significantly reduce the reaction temperature of PET methanolysis, but the use of a large number of cosolvents not only causes increased energy and carbon emissions, but also produces large amounts of waste liquid. Besides, a large amount of methanol is used in common work which far exceed the equivalent methanol required for PET degradation (the mass ratio is 0.33: 1), which further causes waste and increases reaction pressure. In these regards, it is crucial to develop a new catalyst to achieve efficient degradation of waste PET at low temperatures with low usage of methanol.

Deep eutectic solvents (DESSs) could form a large number of hydrogen bonds with reactants through the selection of HBDs and HBAs (Rente et al., 2022). The strong hydrogen bonding between DES and reactants is beneficial to the activation of carbonyl in PET and hydroxyl in methanol (Li et al., 2024). At the same time, the strong interaction can reduce the amount of methanol, thus increasing the concentration of catalyst in the reaction system, which is expected to realize the methanol alcoholysis of

PET at lower temperature. Moreover, the relatively mild reaction conditions can prevent the esterification of DMT to MHET and BHET, thereby increasing the yield of DMT (Jadhav et al., 2020). This has inspired us to design a DES catalyst with high activity at low temperature, aiming to achieve PET degradation at mild condition while reducing the degree of DMT transesterification, thereby minimizing methanol usage and realizing green energy-saving.

In this work, we developed an effective catalyst, TMG/DBN, to convert PET to DMT with equivalent amount of methanol at low temperature. Although a small amount of methanol increased the concentration of the by-product EG in the reaction system, it has little effect on the transesterification of product DMT, and the yield of DMT was as high as 93.1 %. A series of characterizations and theoretical calculations were studied to explain this phenomenon. The reaction mechanism was further explored. Theoretical calculation was also used to analyze the process of depolymerization.

2. Experimental section

2.1. Materials

PET flakes were purchased from Beijing Incom Resources Recovery Recycling Co., Ltd with an average size of 5 mm × 5 mm. 1,5-diazabicyclo[4.3.0]-5-nonene (DBN, 98 %), phenol (99.5 %), 1,3-bis(hydroxymethyl)urea (DMU, 98 %) N,N'-dimethylurea (98 %) dimethyl terephthalate (DMT, 99 %), dimethyl isophthalate (DMI, 99 %), betaine (98 %) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Methanesulfonic acid (98 %), 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD, 97 %), 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU, 99 %), tetramethylguanidine (TMG, 98 %), tropine(98 %), zinc acetate (99 %) were obtained from Shanghai Macklin Biochemical Co., Ltd. Methanol (AR), acetonitrile (AR) and ethylene glycol (EG, AR) were purchased from Tianjin Damao Chemical Reagent Factory. 2-hydroxyethyl methyl terephthalate (MHET, 97 %) was obtained from Tokyo Chemical Industry Co., Ltd. Bis(hydroxyethyl) terephthalate (BHET) was obtained through laboratory refinement.

2.2. Catalysts preparation

TMG/DBN was synthesized by mixing TMG and DBN at 1:1 (molar ratio) at 80 °C for 5h in a round flask until clear and homogeneous liquid formed, which was dried in a vacuum oven at 60 °C for 12 h. The synthesis of other catalysts was shown in supporting information.

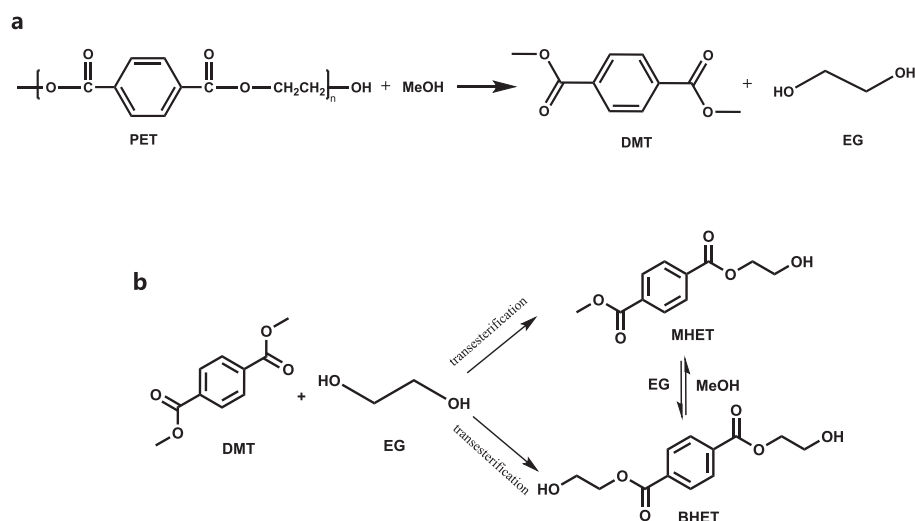


Fig. 1. (a) The process of PET methanolysis, (b) The conversion of DMT to MHET and BHET.

2.3. General procedure of PET methanolysis

For each experiment, 1 g PET flakes (W_1), a suitable quantity of methanol and catalyst were added into a pressure-resistant reaction flask with a magnetic stirring. The reaction was carried out in a magnetic stirring oil bath heater at the temperature range of 95–120 °C with the reaction time range from 40 to 180 min. At the end of the reaction, a certain quantity of acetonitrile was added to it after cooled to room temperature. Unreacted PET flakes were separated by filtration after fully dissolved, and dried at 60 °C for 12 h and weighted (W_2). The conversion of PET is calculated by the following equation:

$$C_{PET} = \frac{W_2 - W_1}{W_1} \times 100\% \quad (1)$$

where W_1 is the initial weight of PET and W_2 is the weight of residual PET. The yield of DMT is calculated by Eq. (2) as follow:

$$Y_{DMT} = \frac{W_{DMT}/M_{DMT}}{W_1/M_{PET}} \quad (2)$$

where W_{DMT} is the weight of DMT which is calculated by quantitative analysis of HPLC, M_{DMT} is the molecular weight of DMT, M_{PET} is the repeat unit molecular weight of PET.

The yield of MHET is calculated by Eq. (3) as follow:

$$Y_{MHET} = \frac{W_{MHET}/M_{MHET}}{W_1/M_{PET}} \quad (3)$$

where W_{MHET} is the weight of MHET, M_{MHET} is the molecular weight of MHET.

The yield of BHET is calculated by Eq. (4) as follow:

$$Y_{BHET} = \frac{W_{BHET}/M_{BHET}}{W_1/M_{PET}} \quad (4)$$

where W_{BHET} is the weight of BHET, M_{BHET} is the molecular weight of BHET.

The yield of DMI is calculated by Eq. (5) as follow:

$$Y_{DMI} = \frac{W_{DMI}/M_{DMI}}{W_1/M_{PET}} \quad (5)$$

where W_{DMI} is the weight of DMI, M_{DMI} is the molecular weight of DMI.

The obtained filtrate was filtered using PTFE 0.2 μm filters and analyzed using high-performance liquid chromatography (HPLC, LC-20AT, Shimadzu, Japan) equipped with refractive index (RI) and photo-diode array (PDA) detectors. The HPLC utilized a Shimadzu InertSustain AQ-C18 column (5 μm , columns 4.6 $\times$ 250 mm) and a mixture solution of methanol and ultrapure water (70: 30 (v/v)) as a mobile phase flowing at 0.5 $\text{ml}\cdot\text{min}^{-1}$ and 35 °C to perform the separation.

2.4. Characterization

The ^1H NMR spectrum of the catalyst was obtained by a nuclear magnetic resonance (NMR) instrument (Bruker, Germany) at 700 MHz in d_6 -DMSO solvent. The FT-IR structures of catalyst and main product were tested by Nicolet 380 spectrometer from Thermo Fisher Scientific. Thermogravimetric analysis (TGA) curves of catalysts were acquired using TGA-50H (Japan) by heating the sample from room temperature to 400 °C at a rate of 10 °C/min under a nitrogen environment. The initial PET and residual PET morphologies were characterized using a scanning electron microscope (SEM, Hitachi, SU8020, Japan). The molecular weights of the initial PET and residual oligomers were determined by gel permeation chromatography (GPC, Agilent PL-GPC 50, USA) with a refractive index (RI) detector and three columns (PLgel 5 μm guard, 50 $\times$ 7.5 mm; PLgel 5 μm MIXED-D, 300 $\times$ 7.5 mm; PLgel 5 μm MIXED-C, 300 $\times$ 7.5 mm). The column oven temperature was 40 °C, the

mobile phase was hexafluoroisopropanol, and the flow rate was 0.5 mL/min . The GPC was calibrated and standard curves were established using standard poly(methyl methacrylate) samples. Electrospray ionization mass spectrometry (ESI-MS) spectra were performed on UPLC-QTOF instrument (Bruker, Germany).

2.5. Computational details

All calculations were implemented in Materials Studio 6.0 with the DMol³ code (Delley, 2000, 1990). The Perdew-Burke-Ernzerhof (PBE) functional of the generalized gradient approximation (GGA) was utilized to calculate the exchange–correlation energy (Perdew et al., 1996). For accurate electronic convergence, double numerical plus polarization (DNP) was used as the atomic orbital basis set in the computations, with a smearing value of 0.005 Å. To accomplish the accuracy and efficiency of computations, effective core potential (ECP) was used. The DFT-D method of Grimme was employed to describe the van der Waals interactions. COSMO solvation model was considered to obtain more accurately results, with methanol as the solvent and dielectric constant was 32.63. The convergence tolerances of energy change, maximum force, and maximum displacement were set as 2×10^{-5} Ha, 0.004 Ha/Å, and 0.005 Å, respectively.

The Gibbs free energy was calculated as:

$$G = E_{elec} + E_{ZPE} + \int C_p dT - TS \quad (6)$$

where E_{elec} is the electronic energy calculated by DFT; E_{ZPE} is the zero point energy (ZPE), C_p is heat capacity, T is temperature, and S is entropy.

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3. Results and discussion

As shown in Fig. S1, a screening of catalysts for the methanolysis of PET such as Tropine/MSA, Tropine/DBN, Tropine/DBU, Tropine/TMG, TMG/DMU, TMG/DBU, TMG/DBN, TMG/TBD were conducted. The selected catalyst with better effect is TMG/DBN.

3.1. Characterization of catalyst

The catalyst was characterized by FT-IR and NMR. The FT-IR spectra are shown in Fig. 2(a). It shows that the –NH peak of TMG in the TMG/DBN catalyst shifted from 3327.40 cm^{-1} to 3330.27 cm^{-1} , which can be ascribed to the formation of hydrogen bond between DBN and TMG. The ^1H NMR spectra are shown in Fig. 2(b). The chemical shift of –NH of TMG, from 3.36 ppm to 4.05 ppm, indicates the formation of hydrogen bond when TMG/DBN catalyst formed. TGA analysis of the catalysts was also studied and the results showed that the stability of the synthesized TMG/DBN catalyst was better than that of TMG but worse than that of DBN. This may be due to the stronger alkalinity of the synthesized catalyst, while the superbase-derived eutectic solvents (ESs) still has volatility even at room temperature and atmospheric pressure (Chen et al., 2021). In the TGA analysis, DES samples were heated from ambient temperatures to 400 °C at a heating rate of 10 °C/min with flowing nitrogen, it is an open system, so it is speculated that the weight reduction of TMG/DBN is due to volatilization. However, the PET methanolysis system is a closed system during the reaction process, it could avoid the loss of TMG/DBN catalyst, and the catalyst is still functioning in the system.

3.2. Characterization of degradation products

A number of characterization methods, including NMR, FT-IR, and

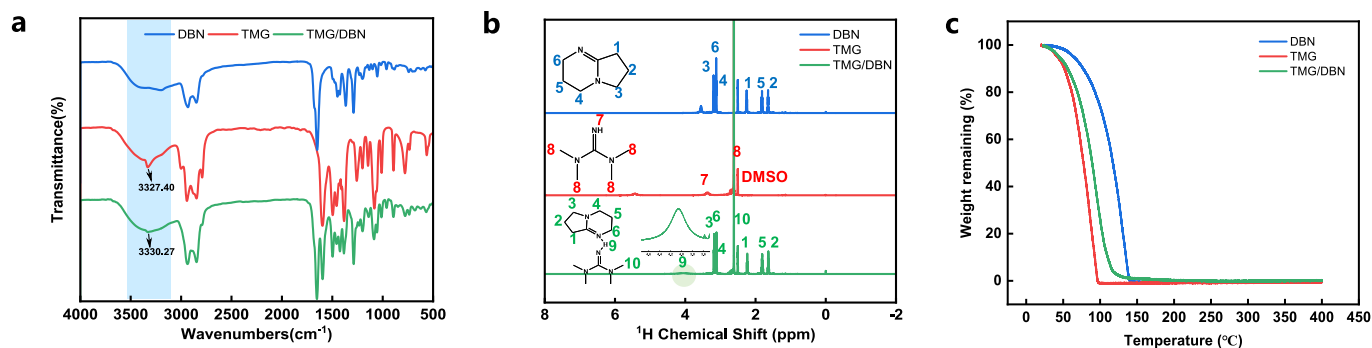


Fig. 2. (a) FT-IR spectra of catalysts, (b) ¹H NMR spectra of DBN (blue), TMG (red), TMG/DBN (green). (c) TGA analysis of DBN (blue), TMG (red), TMG/DBN (green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

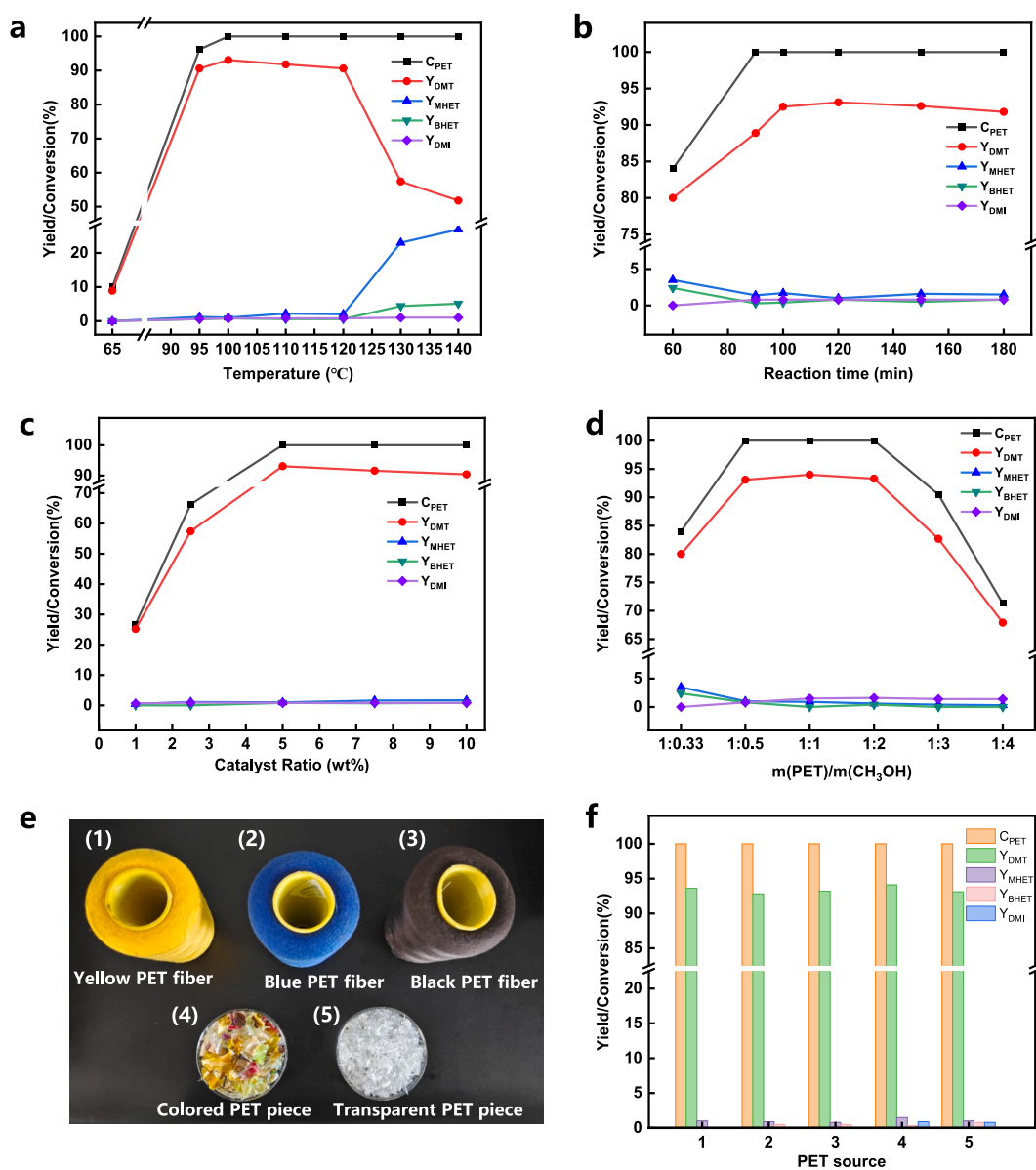


Fig. 3. (a) Effect of reaction temperature on methanolysis process, (b) effect of reaction time on methanolysis process, (c) effect of catalyst amount on methanolysis process, (d) effect of m(CH₃OH)/m(PET) on methanolysis process, (e) commercial PET samples, (f) effect of PET source on methanolysis of PET. If not specified, the following reaction parameters were used by default: reaction temperature (100 °C), reaction time (120 min), catalyst (5 wt% of PET), m(PET): m(CH₃OH) = 1: 0.5.

ESI-MS, were used to ascertain the structure of the product. Fig. S2 displays the results. The peak at $m/z = 195$ in the ESI-MS spectrum (Fig. S2(a)) has the maximum intensity, which is in line with the molecular weight of the DMT ionized by H^+ . The FT-IR spectra of the acquired DMT and standard DMT are shown in Fig. S2(b). It is evident that the hydroxyl fraction is represented by the bands at 3421 cm^{-1} . Stretching band of $C=O$ appears at 1721 cm^{-1} , alkyl C-H bands was at 2959 cm^{-1} , 1439 cm^{-1} was aromatic C-H bands, all of which are present in DMT. The 1H NMR spectrum and ^{13}C NMR spectrum of obtained DMT were shown in Fig. S2(c) and (d), the signals are consistent with standard DMT. High-purity DMT is thus the product of PET methanolysis.

3.3. Investigation of methanolysis process

The catalytic activity was explored with the indicator of efficiency for methanolysis of PET. Fig. 3(a-d) illustrates the results of a thorough investigation into the effects of reaction temperature, reaction time, catalyst amount and usage of methanol on the methanolysis of PET and DMT production. The evolutions of PET conversion and product yields with reaction temperature is depicted in Fig. 3(a). It is noted that, the reaction temperature has a crucial impact on the yield of DMT. When the temperature was 65°C , the conversion of PET and the yield of DMT were only 10 % and 8.9 %, respectively. When the temperature increased to 100°C , the conversion of PET and the yield of DMT reached to 100 % and 93.1 %, respectively. With further increasing temperature, the yield of DMT almost did not increase. In addition, DMT yield was significantly decreased and a large amount of MHET was generated at 130°C , which was related to the reaction of DMT with ethylene glycol (EG) at high temperature. With temperature increasing, the degree of DMT esterification increased (Jadhav et al., 2020) which is different with other studies. It was speculated that the higher reaction temperature leads to more methanol from liquid to gas, reducing the methanol content in the liquid phase and increases the relative concentration of EG in the reaction system.

As shown in Fig. 3(b), the influence of react time was investigated. The conversion of PET and yield of DMT reached the maximum at 120 min. However, the yield of DMT descended lightly with further prolonged reaction time. This result is related to the fact that the methanolysis of PET is an equilibrium reaction. At 120 min, the yield of DMT was the highest, and the reaction basically reached equilibrium in this system. Further prolonging the reaction time, the production of DMT changed lightly.

As shown in Fig. 3(c), when the amount of catalyst increased from 1 % to 2.5 % of PET, the conversion of PET and yield of DMT increased rapidly. This result may be in relation to the increase of catalytically active sites (Jiang et al., 2022). When the catalyst was increased to 5 wt % of PET, the highest DMT yield of 93.1 % was obtained. With the further increase of the catalyst amount, the occurrence of side reactions was aggravated and the yield of the main product was reduced. Therefore, 5 % was the optimized catalyst ratio.

Fig. 3(d) shows the effect of PET/ CH_3OH mass ratio on the degradation of PET. When the mass ratio was 1: 0.5, the conversion of PET was 100 % and yield of DMT was 93.1 %. As the mass ratio was increased to 1: 1 and 1: 2, DMT yield did not increase. When the methanol ratio further reached to 1: 3 and 1: 4, PET was not degraded completely and the yield of DMT decreased. This phenomenon was attributed to a decrease in concentration of catalyst. Considering carbon emission and the cost of industrial production, the most suitable PET/ CH_3OH mass ratio is 1: 0.5. Interestingly, when the mass ratio of PET/ CH_3OH was decreased to 1: 0.33 (equivalent methanol required for PET degradation), despite insufficient reaction, DMT yield reached 80.0 % and the amount of generated MHET was still small (less than 5 %). The result is different from the previous study (Li et al., 2023), where an 80 % yield of DMT was obtained with the PET/ CH_3OH mass ratio of 1: 1, and approximately 15 % by-products of MHET and BHET were produced during the reaction. In other words, under the reaction condition, the

formation of by-product EG has little effect on the reaction. The specific reason will be analyzed in Part 3.5.

In brief, the optimized reaction conditions are as follows: reaction temperature 100°C , reaction time 120 min, catalyst dosage 5 wt% of PET, $m(\text{CH}_3\text{OH}): m(\text{PET}) = 0.5: 1$. The PET conversion rate was 100 % and the yield of DMT was 93.1 % under optimized reaction conditions. The regenerated DMT is also be determined as Fig. S2 shown.

The influence of PET source was also investigated at the same time. As shown in Fig. 3(f), the colored PET piece and fibers were degraded at the optimized reaction conditions. The results reflected that the source of PET and the impurities in it did not affect the activity of the catalyst for the degradation process, indicating that the catalyst has a wide range of applicability.

In addition, a comparison of the catalytic performance of TMG/DBN with other metal salts and ILs is shown in Fig. 4. The reaction temperature of PET methanolysis reported in the literature usually ranges from 130°C to 200°C , and a large amount of MeOH was consumed, from 4 times to more than 8 times the weight of PET (Du et al., 2020; Jiang et al., 2022; Laldinpuui et al., 2022; Tang et al., 2022; Ye et al., 2023). In this work, the nonmetallic catalyst TMG/DBN produced high yield of DMT at a lower temperature with methanol amount far lower than other works used, making the process greener as illustrated in supporting information, Table S4.

3.4. Catalytic kinetics of PET methanolysis

According to published reports, the PET methanolysis reaction order was found to be first order (Al-Sabagh et al., 2014; Chen et al., 2013; Zhou et al., 2019). According to this, the kinetic analysis was performed based on first order initially, and the kinetic equation is as follows:

$$\ln \frac{1}{1-X} = kt \quad (7)$$

Where X is the PET conversion, t is the reaction time and k is the reaction rate constant.

The results of linear regression equation after fitting each group of reaction temperature are shown in Table S1. Using the following Arrhenius equation and the experimental data, the activation energy (E_a) and pre-exponential factor (A) were determined:

$$\ln k = -\frac{E_a}{RT} + \ln A \quad (8)$$

where k is the rate constant, E_a is the activation energy, A is the pre-exponential factor, R is the ideal gas constant, and T is the temperature in Kelvin.

The relationship of reaction time and temperature was depicted in Fig. 5(a), and the results showed a great linear relationship which corroborated this reaction was first-order kinetic model. According to the reaction rate constants, the Arrhenius plot was generated, as shown in Fig. 5(b). Based on the slope of the Arrhenius plot, the E_a was determined to be 84.9 kJ mol^{-1} , which was significantly less than the prior value reported in the literature (ca. 107.6 kJ/mol) (Jiang et al., 2022; Li et al., 2024). According to reports, a reaction's activation energy can vary depending on a number of parameters, including the kind of catalyst, the reaction solvents, and the heating technique. Good catalytic performance of catalyst can reduce the activation energy of the reaction (van Putten et al., 2013).

3.5. Exploration for high selectivity of catalysts

The by-product EG is the important indicator affecting the production of MHET and BHET during the methanolysis of PET as mentioned before. In 3.2, the influence of PET/ CH_3OH mass ratio was researched. In this section, EG is added into the reaction system at the initiation of the reaction. Fig. 6(a) and (b) presents the impact of EG on the yield of

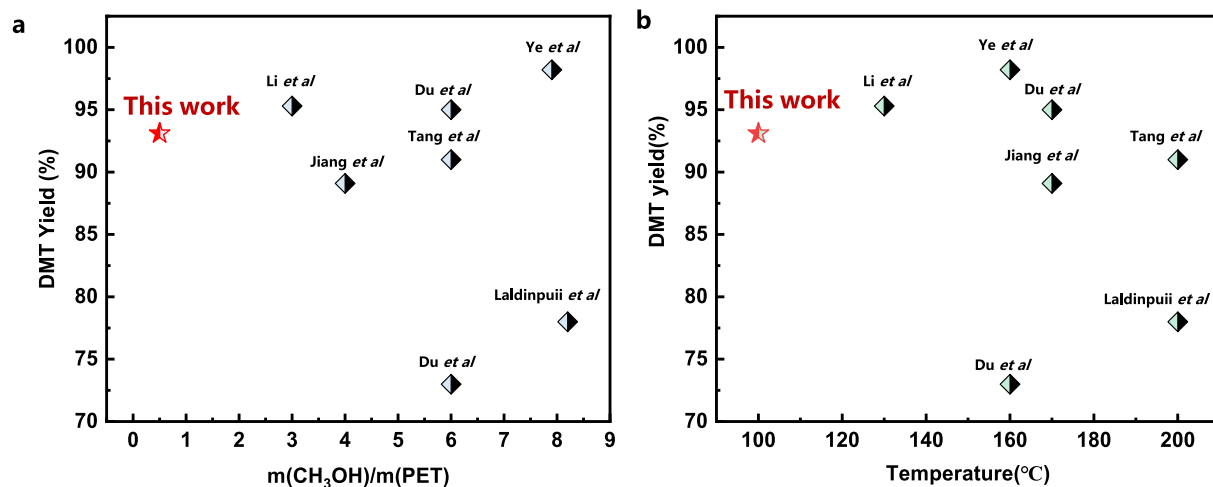


Fig. 4. Comparative data collected from literature for PET methanolysis with other catalysts: (a) DMT yield vs PET/CH₃OH mass ratio and (b) temperature vs DMT yield.

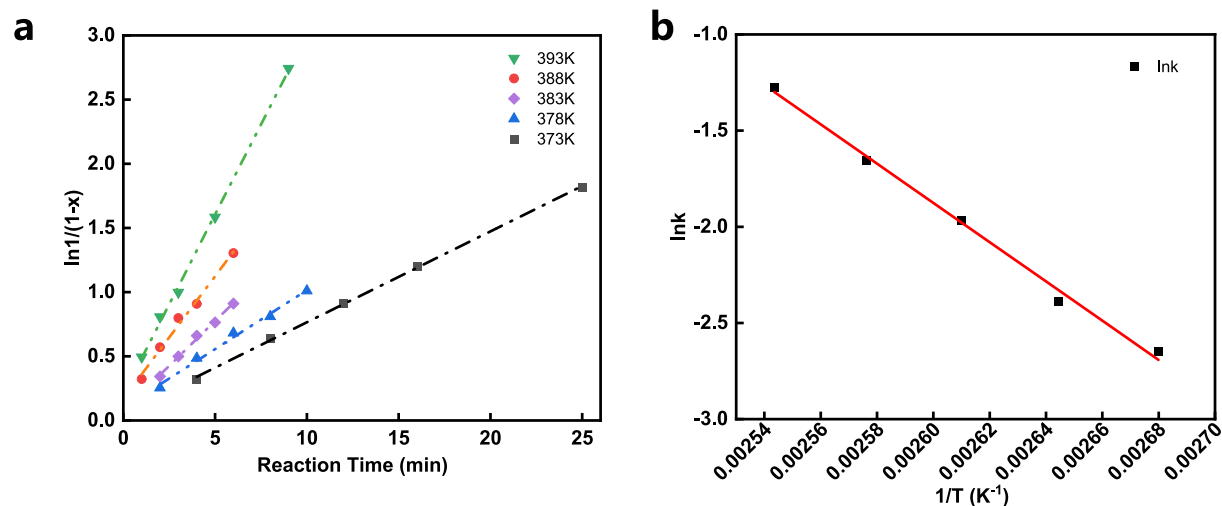


Fig. 5. Catalytic kinetics of PET methanolysis: (a) Effect of temperature on the rate of PET methanolysis. Reaction conditions: PET, 1 g; MeOH, 0.5 g; TMG/DBN, 0.05 g. (b) Arrhenius plot of the rate constant of PET methanolysis.

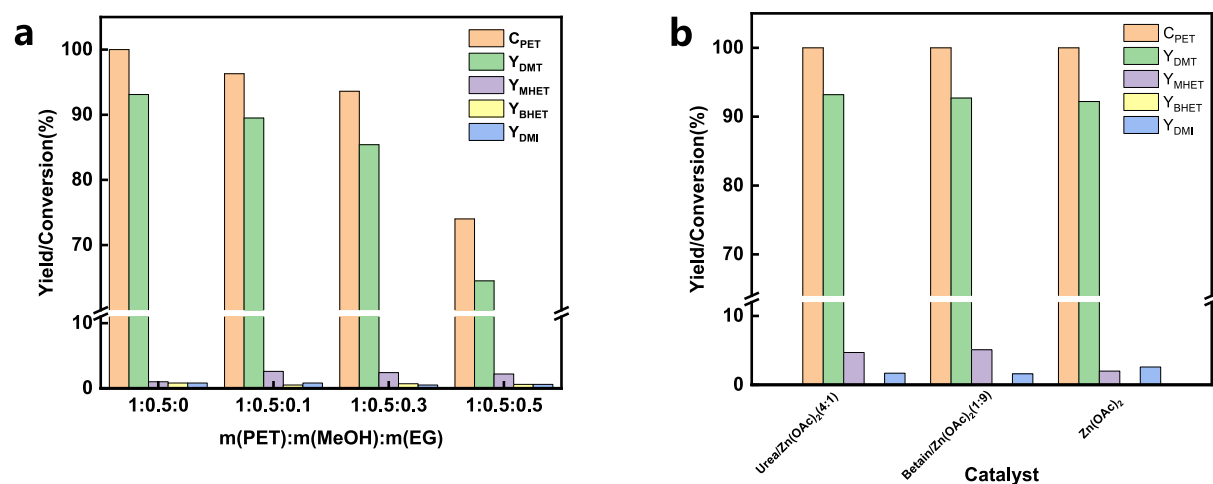


Fig. 6. (a) Effect of EG content on PET methanolysis, (b) PET methanolysis using metal-based catalysts under the conditions: 170 °C, 60 min, catalyst usage 0.005 g, PET 1 g, methanol 4 g.

DMT and others by-product at 100 °C. When 0.1 g EG was added into the system under the optimized reaction conditions, although PET was not completely degraded, the DMT yield only decreased by about 4 %. When the mass ratio of added MeOH and EG was 0.5: 0.3, which meant about 78 % EG was presented in the solvent at the end of reaction, the conversion of PET and yield of DMT were 93.6 % and 85.4 %, respectively. When the same amount of methanol and EG were added at the beginning of the reaction, the conversion of PET and yield of DMT reached 70.0 % and 64.5 %, respectively. Only 2.2 % MHET and 0.6 % BHET were generated, which was closely related to by-product EG. The results show that EG content has little influence on methanolysis of PET, and TMG/DBN catalyst has good selectivity at low usage of MeOH.

In order to explain the good selectivity of TMG/DBN, the interaction between catalyst and CH₃OH or catalyst and EG was studied by ¹H NMR. As shown in Fig. 7(a) and 7(b), the characteristic peak of hydroxyl hydrogen in CH₃OH was 4.04 ppm and shifted to 4.43 ppm when it mixed with triple the mass of TMG/DBN, while the hydroxyl hydrogen in EG was observed to move from 4.41 ppm to 4.70 ppm when it mixed with three times the mass of TMG/DBN. This phenomenon confirms a stronger interaction between TMG/DBN and methanol than TMG/DBN and EG, which makes the reaction of DMT with EG to generate MHET and BHET a challenge. Molecular dynamics simulation was carried out to further verify the conclusion. Fig. S5 shows the radial distribution function of the centroid distance between catalyst and methanol and EG, respectively. It is found that compared to EG, methanol is closer to TMG/DBN. In other words, when methanol and EG exist simultaneously, the

favorable spatial positioning of methanol is beneficial to its combination with the catalyst. This interaction, to some extent, obstructs the catalyst's binding with EG, thereby impeding the transesterification process.

Compared with TMG/DBN catalyst, PET methanolysis using metal-based catalysts obtained copious by-products MHET and BHET at the optimized reaction conditions, when the mass ratio of PET to MeOH was 1: 4, in which the by-product EG content was very small, indicating that the selectivity of the catalyst was poor.

In order to clarify the reasons for the difference in catalyst selectivity, ¹H NMR was further used for exploring the interaction between Zn(OAc)₂ catalyst and CH₃OH or EG. As shown in Fig. 7(c) and 7(d), when the mass ratio of Zn(OAc)₂ to CH₃OH reached 3: 1, the peak of the hydroxyl hydrogen in CH₃OH shifted from 4.04 ppm to 4.11 ppm, with a change of 0.07 ppm. This shift is nearly identical to that observed in EG, where the peak moved from 4.41 ppm to 4.47 ppm, resulting in a change of 0.06 ppm. The result indicates that the interaction between Zn(OAc)₂ with methanol and EG is almost the same, which makes it easier for the catalyst to combine with EG. The strong interaction between Zn(OAc)₂ and DMT was characterized by ¹³C NMR (Fig. S4.). The carbon in carbonyl of DMT had a notable chemical shift from 165.50 ppm to 165.56 ppm when Zn(OAc)₂ mixed with DMT. when TMG/DBN was mixed with DMT, the shift value was only 0.02 ppm, indicating there was greater interaction between Zn(OAc)₂ and DMT making transformation of DMT to MHET and BHET more easily to occur.

As shown in Fig. 8(a-c), binding energy was also calculated to prove above results. The binding energy between TMG/DBN and MeOH is

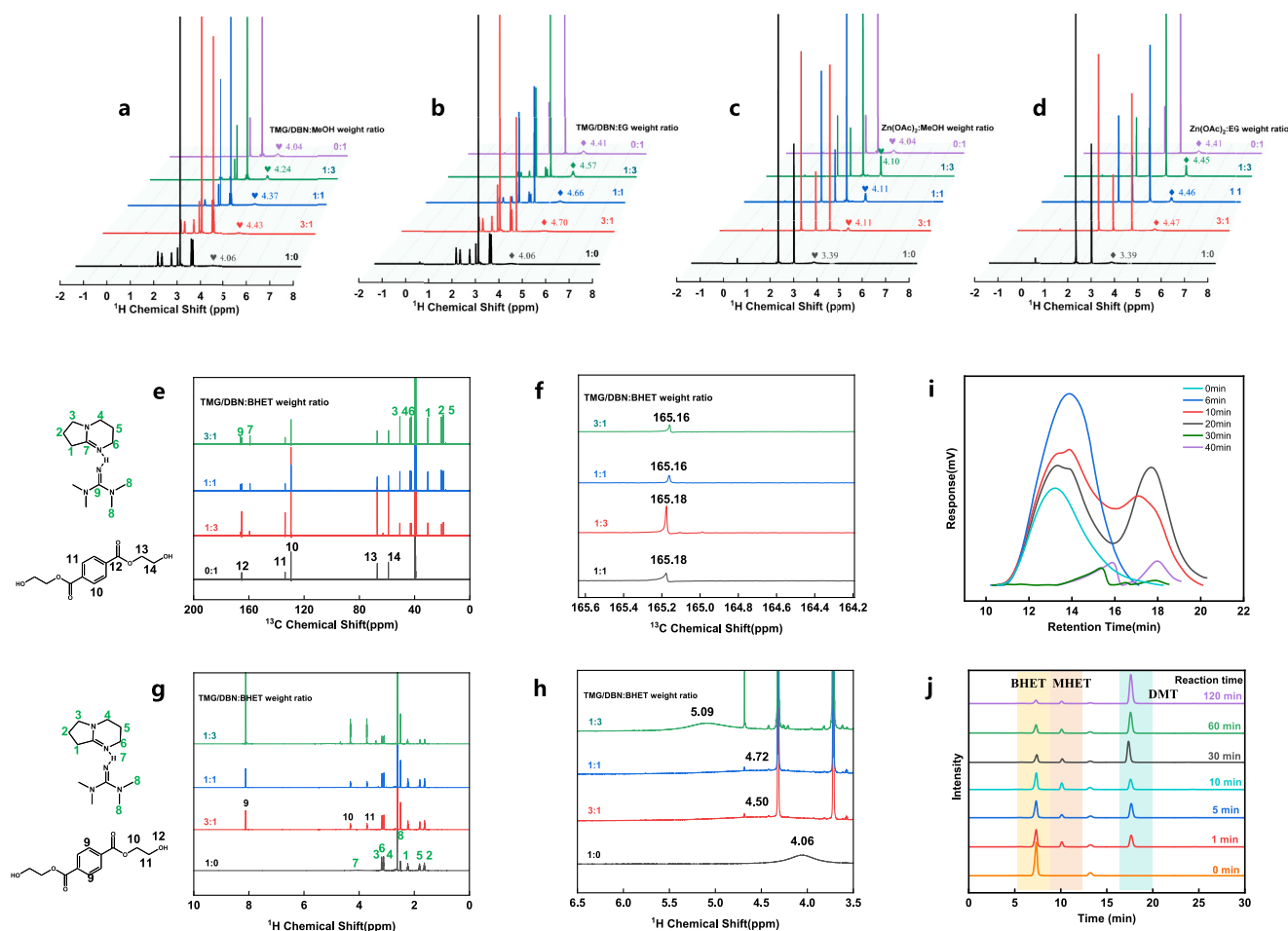


Fig. 7. ¹H NMR spectra of catalyst and solvent mixtures with different weight ratio: (a) TMG/DBN and MeOH, (b) TMG/DBN and EG, (c) Zn(OAc)₂ and MeOH, (d) Zn(OAc)₂ and EG. NMR spectra of catalyst and PET model compound BHET with different weight ratio: (e) ¹³C NMR spectra, (f) signals of ¹³C NMR spectra from 164.2 to 165.6 ppm, (g) ¹H NMR spectra, (h) signals of ¹H NMR spectra from 6.5 to 3.5 ppm. Investigation of intermediate products in the methanolysis of PET: (i) GPC analysis of PET at different reaction times, (j) HPLC analysis of products at different reaction times with BHET as model compound of PET.

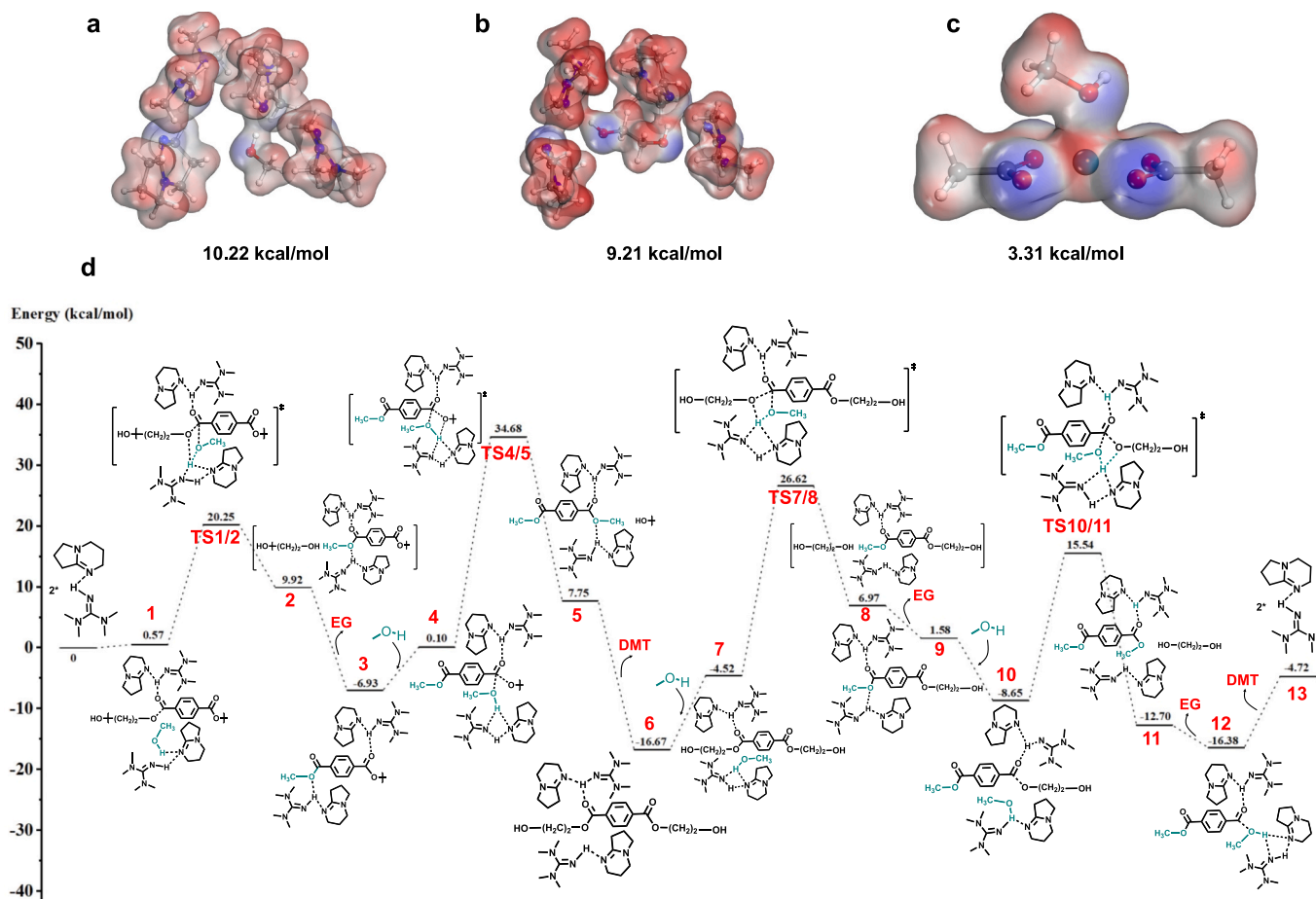


Fig. 8. Binding energy of catalysts with methanol or EG: (a) TMG/DBN and MeOH, (b) TMG/DBN and EG, (c) $\text{Zn}(\text{OAc})_2$ and MeOH. (d) Gibbs free energy profile for methanolysis of PET by TMG/DBN catalyst.

10.22 kcal/mol, which is larger than the energy between DES catalyst and EG (9.21 kcal/mol). Besides, this value is also greater than the binding energy between $\text{Zn}(\text{OAc})_2$ and methanol (3.31 kcal/mol), thereby confirming that a stronger interaction exists between TMG/DBN and methanol.

3.6. Mechanistic investigation of PET methanolysis by TMG/DBN

3.6.1. Interaction between the catalyst and reactants

The FT-IR analyses was shown in Fig. S6, the hydroxyl hydrogen of the mixture of TMG/DBN and CH_3OH with a molar ratio of 1: 1 has a right shift compared with the catalyst and CH_3OH individually which indicates that TMG/DBN and CH_3OH can interact to form hydrogen bonds and activate hydroxyl groups. The interaction between TMG/DBN and CH_3OH was investigated by NMR in 3.5.

The interaction between the catalyst and the PET model compound BHET was explored by ^1H NMR and ^{13}C NMR, as shown in Fig. 7(e-h). The carbon in carbonyl of BHET has a chemical shift when BHET mixed with catalyst. As shown in Fig. 7(g) and (h), the $-\text{NH}$ bond of the catalyst was shifted as well, indicating the interaction between the catalyst and BHET. It is concluded that the $-\text{NH}$ bond in the catalyst could interact with the $\text{C}=\text{O}$ bond in PET, promoting the reactions.

3.6.2. Investigation of intermediate products

The molecular weight of the residual PET flakes at different reaction times were analyzed by gel permeation chromatography (GPC), as shown in Fig. 7(i). When the reaction time was less than 10 min, the PET bottle flakes only exhibited a high molecular weight peak. With the extension of reaction time, a low molecular weight peak appeared.

Besides, during the entire reaction process, with the extension of reaction time, the molecular weight of the remaining PET bottle fragments showed a peak shift towards the low molecular weight. The results indicated that PET is degraded with the reaction progressed, resulting in a decrease in molecular weight. The SEM results also coincide with this suggestion. As the images (Fig. S7 and S8) shows, the surface of PET was damaged and produced cracks on the first layer at the beginning of reaction due to contact with MeOH and catalyst. As the reaction progressed, cracks become larger, more fragments were produced and detached due to the formation of product. Then next layer was also exposed, and the reaction extended from the first layer to the second layer. The surface of the second layer became rough and rifts appeared.

The experiments were taken to observe the distribution of products during the reaction process. Analyzing the yield of product at different reaction times in Fig. 7(d), it was observed that the contents of BHET and DMT reached a stable stage during the whole process. It could be inferred that the transesterification of DMT to BHET took place to some extent and process was believed to be in equilibrium.

In order to prove that, BHET was used as PET model compound, the reaction temperature and catalyst dosage both decreased to observe the reaction process more clearly, the experimental conditions were adjusted as follows: BHET (1 g), CH_3OH (0.5 g), TMG/DBN (0.5 wt% of BHET), and 30°C and the HPLC spectra were shown in Fig. 7(j). The concentration of PET model compound BHET gradually decreases, while the amount of the generated DMT progressively increases during the whole reaction. It is notable that the quantity of the by-product MHET is minimal at the initial stage of the reaction and remains unchanged as the reaction proceeds, which is different with previous work (Li et al., 2024). Besides, BHET could not convert into DMT completely even

though within a long reaction time. Although the content of EG does not have a significant effect on the DMT yield, its presence does lead to the generation of MHET and BHET. This also means that there is a reaction equilibrium between the conversion of BHET and DMT.

3.6.3. Theoretical calculations

The theoretical simulation of possible pathways for the PET methanolysis studied for further exploration of the reaction mechanism. As seen in Fig. 8(b), [BHET]₂ was employed as the model of PET in the calculation. At the beginning of the reaction, the catalyst associates with [BHET]₂ and MeOH and then form complex 1, where the Gibbs reaction energy (ΔG) was 0.57 kcal/mol. The C=O bond at one end of [BHET]₂ was activated by the catalyst, and the C-OR bond is broken to form complex 2, where the Gibbs activation energy was 19.68 kcal/mol. Then EG was released and the catalyst was associate with another MeOH molecular, forming complex 4, which had a new ester bond formed by esterification reaction with methanol. The C=O double bond connect with another BHET was activated by the catalyst, resulting in the cleavage of the C-OR bond and the formation of complex 6, accompanied by the release of one DMT monomer. The Gibbs reaction energy and Gibbs activation energy were -16.1 and 34.11 (reference to complex 1) kcal/mol, respectively.

Then, another methanol molecule was introduced to form complex 7 and produced DMT monomer. The Gibbs activation energy of forming MHET was 26.05 kcal/mol (reference to complex 1), while the energy of forming DMT was 16.97 kcal/mol (reference to complex 1). This indicated that the activation energy required for the generation of MHET was higher than that for DMT, it is the rate determining step; thus, at a constant temperature, the reaction for generating DMT would be easier to proceed. The aforementioned conclusion can also account for the consistently low level of MHET production depicted in Fig. 7(j) throughout the entire reaction.

When the reaction was finished, the catalyst is released, completing the catalytic cycle. It is also noteworthy that, compared to the activation of the C=O double bond end group and the cleavage of the C-OR bond, the cleavage of the C=O bond in [BHET]₂ requires more energy, potentially due to increased space steric hindrance.

3.6.4. Analysis of catalyst-treated PET

The GPC analysis of original PET, the PET immersed in TMG/DBN and its individual components at 100 °C for 2 h, was presented in Fig. S10. A significant decrease in the molecular weight of residual PET soaked in TMG, DBN and catalyst indicated the degradation of PET, especially in TMG/DBN. It is deduced that TMG/DBN catalyst has a stronger interaction with the C=O bond of PET, which reduces the activation energy and reaction temperature of the methanolysis of PET.

In order to gain more insights into the effect of catalyst, we performed SEM analysis of the PET sample soaked in TMG, DBN and catalyst at 100°C for different time. As the SEM images (Fig. S11 and S12) show, the surface of PET got rough as the immersion time of TMG/DBN increased. Compared with PET soaked in TMG and DBN individually for 2 h, the TMG/DBN-immersed PET surface became rough, which might enhance the surface area in contact with the catalyst and CH₃OH. After soaked in TMG, small pores appeared on the surface of PET, and the number of pores increased with reaction time. These pores may increase the contact area between the catalyst, methanol and PET and lead to more erosion of the PET surface by TMG/DBN.

In summary, a PET degradation mechanism is proposed as displayed in Fig. S13. The hydrogen atom on the -NH of TMG forms a hydrogen bond with the nitrogen atom on DBN, resulting in an increase in the positive charge on the hydrogen atom. This makes it possible to interact with the oxygen atom on the PET carbonyl group, which raises the positive charge on the carbon atom. Simultaneously, new hydrogen bond is formed between the N atoms of TMG/DBN and the hydroxyl hydrogen of methanol, increasing the negative charge on hydroxyl oxygen, promoted a reaction with the carbon cation on the C=O bond to

activate the ester bonds on PET and form new ester bonds. PET degradation occurs from the surface, initially generating PET oligomers, followed by the formation of intermediate product MHET, and generated DMT finally.

4. Conclusion

In this study, an efficient deep eutectic solvent TMG/DBN was developed and employed in methanolysis of PET at mild reaction condition with low usage of methanol. With this catalyst, the waste PET flakes were depolymerized at 100 °C for 2 h using methanol with half the quality of PET, which is nearly equivalent methanol required for PET degradation. As a result, this process was carried out with 100 % PET conversion and 93.1 % DMT yield. By the study of NMR, FT-IR and theoretical calculation it was revealed that the stronger interaction between the catalyst and methanol than EG plays an important role in inhibiting DMT transesterification. Therefore, the catalyst exhibits excellent catalytic performance with minimal methanol usage and low temperature in the process of PET methanolysis. This further contributes to the reduction of resource waste and plastic pollution, and minimizes substantial carbon emissions resulting from the excessive use of solvents.

Overall, this work provides a useful method for the design and synthesis of catalysts, aiming at degrading waste PET at energy-efficient and economic recycling way.

CCRediT authorship contribution statement

Chunrui Rong: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Dongxia Yan:** Writing – review & editing, Supervision, Investigation. **Jiabao Li:** Investigation, Formal analysis. **Jian Feng:** Writing – review & editing, Investigation. **Jiayu Xin:** Writing – review & editing, Resources. **Qing Zhou:** Writing – review & editing, Resources. **Yi Li:** Resources. **Junli Xu:** Resources. **Xingmei Lu:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ces.2025.121859>.

Data availability

The authors do not have permission to share data.

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